MILITARY SPECIFICATION

GUANIDINE NITRATE (GN)

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions and packaging of Guanidine Nitrate for use in the manufacture of Military Explosives.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

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FSC: 6810

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-129	- Marking for Shipping and Storage
MIL-STD-650	- Explosive: Sampling, Inspection and Testing
MIL-STD-1218	- ACS Chemicals

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E300	Sampling Industrial Chemicals
ASTM-D2905	Statements of Number of Specimens Required to Determine the Average Quality of a Textile Material

CODE OF FEDERAL REGULATIONS

49CFR100-199	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite, "49CFR100-199 (latest revision).)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19013).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Properties. The Guanidine Nitrate (GN) shall conform to the requirements listed in the following table when tested in accordance with the applicable paragraphs:

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complement of fuel, lubricants and coolant; and an operator

TABLE I
REQUIREMENTS FOR GUANIDINE NITRATE

Properties	Requirements	Test Paragraph
Total Volatiles, Percent %, Maximum (Max)	0.8	4.6.1
Purity, %, Miniumum	96.0	4.6.2
Ammonium Nitrate, %, Max.	1.8	4.6.3
Melamine, %, Max.	0.2	4.6.4
Ash %, Max.	0.5	4.6.5
Sulfate as Na_2SO_4, %, Max.	0.1	4.6.6
Free Ammonia, %, Max.	0.5	4.6.7
Water Insolubles, %, Max.	0.2	4.6.8
Temperature Rise ^{o}C, Max.	36	4.6.9

3.2 Granulation. The GN shall be a white, free flowing material which is free from dirt, oil, grease, or other visible impurity and shall conform to the requirements listed in the following Table II when tested as specified in 4.6.10.

TABLE II
GRANULATION REQUIREMENTS

U.S. STANDARD SIEVE NUMBER	AMOUNT PASSED THROUGH SIEVE
16	95 percent, minimum

3.3 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

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a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 200 grams of Guanidine Nitrate obtained by sampling as described in 4.5.1 . The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures equipment and material which will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspection to be performed. The sample will be subjected by the Government to any or all of the examinations for tests specified in 4.6 of this specification.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of one or more batches of GN produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of Guanidine Nitrate that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous (See MIL-A-48078).

4.5 Testing. PRECAUTION: This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Sampling. Two (2) individual representative 200 gram samples from each batch or lot of the GN shall be obtained as specified in ASTM E300, section 25. Each sample shall be tested independently in accordance with the applicable methods given in 4.6. If either sample fails to meet any of the test requirements, the batch or lot shall be rejected. The samples shall be subjected to the tests listed below and MIL-STD-105 shall apply for the defect classification given.

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in accordance with SAE AIR or ... prediction calculations shall be determined

Inspection/Requirement	Defect Classification
Properties (See 3.1)	Major
Granulation (See 3.2)	Major

4.5.2 Inspection equipment. For the performance of all tests and examinations specified in 4.6, commercial inspection equipment should be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes. (See 6.2)

4.6 Test methods and procedures. The tests to determine compliance with the requirements given in Table I shall be performed on the Guanidine Nitrate samples obtained in 4.5.1. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall be Reagent Grade or ACS Grade in accordance with MIL-STD-1218.

4.6.1 Total volatiles. Transfer to a tared glass-stoppered weighing bottle (or a tared aluminum can with a lid) an accurately weighed portion of approximately 2 grams (g) of the sample. With the stopper off, place the sample, weighing bottle and stopper in an oven at 100° to 105° Centigrade (C) for 3 hours, remove and cool in a desiccator, and weigh. Calculate the loss in weight as percent total volatiles as follows:

$$\text{Percent Total Volatiles} = \frac{A-B\ (100)}{A}$$

A = Weight of specimen before drying, g
B = Weight of specimen after drying, g

4.6.2 Purity.

4.6.2.1 Preparation of precipitation/washing solution. Add 7.5g of reagent grade picric acid, 0.075g of reagent grade guanidine nitrate, and 8 milliliters (mL) of concentrated ammonium hydroxide to a 1000 mL volumetric flask. Add 500 mL of distilled water to the flask and mix thoroughly. Fill the flask with distilled water to the calibration mark and allow to stand for approximately 10 days before use.

4.6.2.2 Standardization. Weigh 0.2g of reagent grade guanidine nitrate into a 250 mL erlenmeyer flask. Add 15 mL distilled water and 5-10 drops concentrated ammonium hydroxide. Heat the flask and contents on a hot plate until the guanidine nitrate dissolves. Remove the flask from the hot plate and allow it to cool to room temperature and filter through a No. 42 Whatman

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Filter Paper. To the filtrate slowly add 100 mL of the precipitation solution prepared in 4.6.2.1 to the guanidine nitrate solution while continually swirling the flask. Allow the solution to stand 8-12 hours at room temperature, then filter through a Selas porcelain filtering crucible, of fine porosity, which is attached to a vacuum system containing a crucible holder and filtering flask. Rinse the erlenmeyer flask with portions of the precipitating/washing solution. Wash the precipitate with the washings in the flask. Evacuate the crucible to dryness. Dry the filtering crucible and contents to a constant weight in a drying oven at 105 ± 5°C. Cool the filtering crucible and contents in a desiccator and weigh. Calculate the value of the guanidine nitrate to guanidine picrate factor (A) as follows:

$$A = \frac{(C)\ D}{100\ B}$$

Where:
A - guanidine nitrate to guanidine picrate factor.
B - weight of precipitate in crucible g.
C - weight of sample used g.
D - purity of guanidine nitrate reagent in percent

4.6.2.3 Guanidine nitrate (purity) determination. Repeat the procedure described in 4.6.2.2 using the guanidine nitrate sample instead of the reagent grade guanidine nitrate. Calculate the percent guanidine nitrate as follows:

$$\text{Percent guanidine nitrate} = \frac{(E)\ (A)\ 100}{W}$$

Where:
E - weight of precipitate in crucible (g)
W - weight of sample used (g)
A - guanidine nitrate to guanidine picrate factor determined in 4.6.2.2

4.6.3 Ammonium nitrate.

4.6.3.1 Preparation of a standardized formaldehyde solution. To 25 mL of formaldehyde, add a few drops of phenolphthalein. Slowly titrate the solution with 0.5N or 0.33N sodium hydroxide to a faint pink color.

4.6.3.2 Procedure. Add an accurately weighed portion of the sample, approximately 10g, to a 400mL beaker containing 200mL water. Heat the contents of the beaker on a hot plate and stir the solution until all the GN dissolves. To the beaker and contents add 5.0Ml of the standardized formaldehyde, then 2-3

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drops of phenolphthalein solution. Titrate the solution with 0.5N or 0.33N sodium hydroxide to a faint pink color. Calculate the percent ammonium nitrate as follows:

$$\text{Percent ammonium nitrate} = \frac{(A)(N)(M)\ 100}{1000\ W}$$

Where:
A - mL of sodium hydroxide used in titration.
N - normality (N) of sodium hydroxide.
M - molecular weight of ammonium nitrate.
W - weight of guanidine nitrate sample g.

4.6.4 Melamine. To a 250mL volumetric flask, containing 200mL hot water (70 - 80°C), quantitatively transfer an accurately weighed portion of the sample, approximately 10g. Mix the contents of the flask until sample dissolves completely. Allow the contents of the flask to cool to room temperature and fill the flask to the mark with distilled water and mix. Pipette a 50mL portion of the solution into a 250mL beaker and perform a potentiometric titration on the solution using 1N sulfuric acid and a calomel/antimony electrode. The potential shift occurs at a pH of approximately 3.5 (190 - 200 millivolts). Calculate the percent melamine as follows:

$$\text{Percent melamine} = \frac{A\ (N)(M)\ 100}{100\ P}$$

Where:
A - mL of sulfuric acid used in potentiometric determination.
N - normality (N) of the sulfuric acid.
M - molecular weight of melamine (126).
P - weight in grams of guanidine nitrate introduced in the potentiometric determination. This value may be calculated as follows:

$$P = \frac{(A)\ (B)}{C}$$

Where:
A = weight of GN dissolved, g.
B = volume of solution used in analysis, mL.
C = capacity of volumetric flask, mL.
P = 2.0 when sample size, flask size, and solution volume drawn with pipette are exactly as detailed in the procedure.

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4.6.5 Ash. Weigh accurately approximately a 5g portion of the GN sample into a tared porcelain crucible which was properly cleaned and dried. Carefully and slowly heat the crucible until sample decomposes. Continue heating in an electric furnace (900 ± 50°C) until constant weight is obtained. Calculate the percentage of ash using the following formula:

$$\text{Percent ash} = \frac{(A)\ 100}{W}$$

Where:
A - Weight of contents after ignition, g
W - weight of guanidine nitrate sample g

4.6.6 Sulfate (as sodium sulfate).

4.6.6.1 Materials and equipment.

Hydrochloric acid, concentrated
Barium chloride solution, 10% (by weight)
Silver nitrate solution, 1%
Distilled water at 50°C ± 5°C
Fritted Glass crucible, fine porosity
Beaker, 400 mL
Stirrer, glass
Filter paper, Whatman No. 42
Hot plate
Oven
Muffle furnace
Desiccator

4.6.6.2 Procedure. Transfer an accurately weighed portion of the GN sample, approximately 5.0g, to a 400mL beaker containing 200mL of hot water (80 - 90°C). Use a glass rod and stir to dissolve the sample. If necessary, filter the solution through a No. 42 Whatman filter paper. Acidify the solution by adding 1.0mL of the concentrated hydrochloric acid. Then add 1.0mL of the barium chloride solution. Boil the resultant solution for approximately five minutes on a hot plate, remove from the hot plate, allow the solution to cool to room temperature with occasional gentle stirring. Allow the beaker and contents to remain at room temperature for a minimum of 10 hours. Then, heat to boiling on a hot plate and quantitatively filter the contents of the beaker through a previously ignited and tared Gooch crucible using warm 20mL portions of water to aid in the transfer.

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Wash the crucible and precipitate with 20mL portions of warm water. Draw each portion of the warm water through before adding the next 20mL portion of warm water. Discontinue the washings when the filtrate shows no chlorides when tested with silver nitrate. Place the crucible and contents for one hour in an oven maintained at 100 ± 5°C. Then place the crucible and contents in a muffle furnace (600±25°C) for one hour. Remove crucible and contents from furnace, cool in a desiccator and weigh. Calculate the percent sodium sulfate as follows:

$$\text{Percent sodium sulfate} = \frac{60.86A}{W}$$

Where:
A = Weight of residue, g
W = Weight of sample, g

4.6.7 Free ammonia.

4.6.7.1 Material and equipment

Material
Distilled water
Hydrochloric acid solution, 0.33N
Methyl red indicator solution, 0.1% by weight in isopropyl alcohol

Equipment
Analytical balance, able to weigh to 0.1 mg
Beakers, 250 mL

4.6.7.2 Procedure.

Accurately weigh approximately a 1.0g portion of the sample into a 250mL beaker. Add approximately 100mL of distilled water and add 4 drops of methyl red indicator. If the solution turns yellow, using magnetic stirring, titrate the solution with 0.33N HCl until a red color end point is obtained. Record the number of mL of HCl as A.

Calculate the free ammonia as:

$$\text{Percent Free Ammonia} = \frac{(1.703)\ (A)\ (N \text{ of HCl})}{\text{wt. of sample}}$$

4.6.8 Water insoluble.

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4.6.8.1 Materials and equipment.

Distilled water
Sintered glass crucibles, fine porosity
Oven
Desiccator

4.6.8.2 Procedure. Transfer an accurately weighed portion of the sample, approximately 10g, to a 800mL beaker containing 400mL warm water (50°-75°C). Stir to dissolve sample. Quantitatively transfer the contents of the beaker thru a tared, fine porosity sintered glass crucible. Wash the crucible and contents with five (5) 20mL portions of warm water. Allow each portion of the wash water to go thru the crucible before adding the next portion. Dry the crucible and contents in an oven, maintained at 105±5°C for two hours, then cool in a desiccator and weigh. Calculate the water insolubles as follows:

$$\text{Percent water insoluble impurities} = \frac{(A-B)\ 100}{W}$$

Where:
A = Final weight of crucible, g
B = Initial (tare) weight of crucible, g
W = Weight of sample, g

4.6.9 Temperature rise.

4.6.9.1 Materials and equipment.

98% Sulfuric Acid (technical grade).
Dewar Flask, 500 mL
Stirring rod
Nitration rack
Thermometer, °C
Triple beam balance, 2000 g. capacity

4.6.9.2 Procedure.

Set up the nitration rack over a container of cold water into which the reaction mixture can be dumped and quenched with cold water if the temperature rise is excessive.

Pour 100 mL of sulfuric acid into the Dewar flask in the nitration rack; allow to stand for two minutes, then record the temperature.

NOTE: The temperature should be between 25°C and 30°C

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Add a weighed portion of 18±0.1g of the guanidine nitrate sample into the Dewar flask.

At the end of two minutes, record the temperature; then dispose of the mixture in the approved waste container.

Calculate the temperature rise by subtracting the initial temperature from the final temperature.

A duplicate determination should be made.

4.6.10 Granulation. The granulation of the guanidine nitrate shall be determined in accordance with method 204.1 of Standard MIL-STD-650. Free flowing of the GN shall be determined visually when the GN is poured on the nest of sieves.

5. PACKAGING

5.1 Packing.

5.1.1 Level A. The guanidine nitrate shall be packed in accordance with Title 49 Code of Federal Regulations (CFR) Section 173.182(a)(1).

5.1.2 Level B. The guanidine nitrate shall be packed in accordance with Title 49 CFR Sections 173.182(b)(5) or 173.2(b)(6).

5.1.3 Minimum protection. The guanidine nitrate shall be picked in accordance with any method permitted by Title 49 CFR Section 173.182.

6. NOTES

6.1 Ordering data. See MIL-A-48078.

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

Custodian:
Army-AR

Preparing Activity:
Army-(AR)

Project Number: 6810-AB19

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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